Part A: Conceptual Foundation (Theory)

1. What is Regularization in Machine Learning? Why is it needed?

Regularization adds a penalty to a model's loss function to discourage overly complex models or excessively large coefficients. It helps reduce overfitting and can improve performance on unseen data.

2. Difference between Ridge Regression (L2) and Lasso Regression (L1).

Ridge uses a penalty based on squared coefficients (L2), shrinking coefficients toward zero but usually not making them exactly zero. Lasso uses absolute coefficients (L1) and can set some coefficients to zero, effectively performing feature selection.

3. What is Cross-Validation and why is it important?

Cross-validation splits training data into folds and repeatedly trains and validates a model on different subsets. It gives a more reliable estimate of generalization performance and helps tune hyperparameters without using the final test set.

4. Explain K-Fold, Stratified K-Fold, Leave-One-Out (LOOCV), and Time Series Split.

K-Fold divides data into K folds and rotates the validation fold. Stratified K-Fold preserves class proportions; for regression, target values can be binned to approximate stratification. LOOCV validates on one observation at a time and can be expensive. Time Series Split trains on earlier observations and validates on later observations, preserving chronology.

5. Why are tree-based models less sensitive to feature scaling?

Decision trees split features using threshold comparisons and rely on the ordering of values rather than their magnitudes. Scaling usually preserves that ordering, so tree predictions are generally unaffected. Models such as SVR, Ridge, and Lasso are more sensitive to feature scales.